Applications to probability

1 5 people are to be selected from a larger group of 10 candidates. If Amelia is among the candidates, find the probability that she will be among those selected.

2 5 cards are chosen at random without replacement from a standard deck of 52 cards. Find the probability that they are all red.

3 The nominations for the school captain and vice captain were Ben, Bart, Ellie, Hannah and Emma. The school captain is to be chosen first from these candidates , and then the vice captain. Find the probability that Ben and Ellie will fill the positions.

4 3 letters of the word ENVIOUS are chosen at random. Find the probability that the selection includes just 1 vowel.

5 At a film festival, 6 films are to be chosen to go in the running for grand prize of best film. There are 8 foreign films and 7 local films to choose from. If the selection is made randomly, find the probability that 4 foreign films and 2 local films are chosen.

6 A box contains 6 pens of different colours: red, green, blue, yellow, black and white. Two pens are drawn at random without replacement.

a How many possible selections are there?

b Find the probability of drawing the green and black pens.

7 A newspaper editor is deciding which of 5 articles to print on the front page.

a If she can only choose 2 of them for the front page, how many different selections are possible?

b If Jack wrote one of the 5 articles, find the probability that his article is chosen for the front page.

8 A menu has three entrees (E_1, E_2, E_3), four mains (M_1, M_2, M_3, M_4) and two desserts (D_1, D_2). A meal is made up of one of each.

a How many different meals are possible?

b Find the probability of selecting E_1, M_3 and D_2.

c How many different meals are possible given that E_1 is the entree?

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9 A container contains twelve cards. The cards are marked with the numbers 1 to 12. From the container, five cards are drawn at random without replacement.

- **a** How many possible selections are there?
- **b** Find the probability of selecting 2 as the first number.
- **c** Find the probability of selecting numbers 2, 1 and 4 (in any order) as the first three of the 5 cards.

10 The nominations for the school captain and vice captain were Tom, Fred, Irene, Lisa and Kate. The school captain is chosen first, then the vice captain.

- **a** How many different selections are possible?
- **b** Find the probability of Tom being elected as the school captain and Lisa as the vice captain.

11 In a football squad, there are 9 midfielders. 5 midfielders are selected to play in a game.

- **a** How many selections of 5 midfielders are possible?
- **b** In how many of these selections are Frank and Glen included?
- **c** Find the probability that Frank and Glen will be included in the selection.

12 Companies prefer their product to be placed on the middle shelf in supermarkets. There are 6 brands, but only space for 3 of them on the middle shelf.

- **a** In how many ways can the products for the middle shelf be selected?
- **b** Find the probability that a particular brand is placed on the middle shelf.

13 Senior students are asked to select 8 subjects from any of 7 mainstream subjects and 6 elective subjects.

- **a** How many selections are possible if students can choose any subjects they like?
- **b** Find the probability that a student selects a particular mainstream subject and a particular elective subject among their choices.

14 Four cards of different suits (diamonds, hearts, spades and clubs) are placed in a pile. A card is selected from the pile. Its suit is recorded and the card is returned to the pile. A second card is then chosen and its suit is also recorded.

- **a** How many possible outcomes are there?
- **b** Find the probability of drawing two diamonds.
- **c** Find the probability that the first card is a heart and the second card is a club.
- **d** Find the probability of both cards being black.

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15 A local grocery store offers 26 different flavours of ice-cream. A cone can hold 2 flavours.

a How many selections of two flavours are possible?

b If two of the available flavours are chocolate and vanilla, find the probability of selecting both chocolate and vanilla.

16 You order 12 sandwiches for your office lunch. 5 are roast beef, 5 are ham and 2 are turkey. The sandwich shop wrapped them but forgot to label which meat was in each sandwich. If you pick 4 sandwiches at random, find the probability of the following as a percentage to two decimal places.

a All 4 sandwiches are roast beef.

b Exactly one sandwich is ham.

c 2 are ham and 2 are turkey.

17 4 cards are chosen at random without replacement from a standard deck of 52 cards. Find the following probabilities, as percentages to two decimal places:

a That they are all black.

b That you get two kings and two queens.

c That you get three kings and one random non-king card.

18 Bottles of water are capped by machinery on a production line. After they have been capped, bottles are randomly selected to test that they are correctly sealed. Of the first 20 bottles to come off the production line, 7 of them are not correctly sealed. Georgia randomly selects 5 of the first 20 bottles.

a Find the probability that all of the ones she selected are properly sealed.

b Find the probability that at least one of the bottles she selected is not properly sealed.

19 From 8 teachers and 7 students, 4 people are randomly selected.

a Find the probability that a particular student is chosen.

b Find the probability that there are at least three teachers in the selection.

20 A drawer contains 18 batteries, of which 4 have low charge and the rest are fully charged. Vanessa needs to choose three of the batteries to operate a remote control. Find the probability that:

a All of the batteries she chooses are fully charged.

b At least one of the batteries she chooses has low charge.

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21 An airline wants to allocate 5 flight attendants to a particular flight route. They randomly select them from 8 staff who speak a European language only and 6 staff who speak an Asian language only.

a Find the probability that all the flight attendants speak a European language only.

b Find the probability that there are more Asian than European language speaking attendants.

22 There are 5 red, 5 white and 3 blue marbles in a bag. 5 marbles are chosen at random from the bag, without replacement. Find the probability that:

a The selection consists of all red marbles.

b The selection consists of 4 red and 1 white marble.

c There is at least 1 marble of each colour.

d There are no red marbles.

23 6 cards are chosen from a standard deck of 52 cards, without replacement.

a Find the probability that they are all of the same suit.

b Find the probability that there will be at least one of each suit.

24 A student is choosing two units to study at university: a language and a science unit. They have 4 languages and 7 science units to choose from.

a If they choose one of each, find the total number of combinations of choices.

b If Italian is one of the languages they can choose from, find the probability they choose Italian as their language.

c French is one of the available languages. Find the probability they choose French as their language given that they choose Chemistry as their science unit.

25 A netball coach is choosing players for the Goal Keeper and Goal Defence positions out of the following people:

- **Goal Keeper position**: Beth, Amy, Joy, Tara.
- **Goal Defence position**: Eve, Cara, Daisy, Kim, Liz.

The selection for each position is made independently.

a Find the probability the coach will choose Amy and Daisy.

b Find the probability the coach will choose Amy or Daisy.

c If the coach chooses Joy, find the probability she will choose Kim.

d Find the probability the coach will choose Eve given that Beth won't play with her.

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26 For breakfast each morning, Marge eats porridge, toast or cereal. With that she will either drink orange juice, tea or hot chocolate. Find the probability that Marge will:

- **a** Eat toast and drink tea.
- **b** Eat cereal or drink orange juice.
- **c** Eat porridge given that she drinks hot chocolate.

27 In the Have Sum Fun competition, a teacher needs to make a team of 4 people and another team of 3 people. For the larger team, the teacher has 5 students to choose from, including Jack and Julia. For the smaller team, the teacher has 4 students to choose from, including Alvin. The selections are made independently.

- **a** Find the probability both Jack and Alvin are chosen.
- **b** Find the probability Julia is chosen if Alvin was chosen.
- **c** If Julia won't go without Jack, find the probability they're both on the team.

28 Valentina is creating an exercise plan from a list of 30 exercises. Valentina has a total of 10 cardio exercises, 12 gymnastics exercises and 8 weight exercises. If Valentina wants 6 exercises and one of the chosen exercises is cardio, find the probability that all 6 are cardio.

29 Laura is allowed to pack 4 toys for her weekend trip to Grandma's house. Laura has 6 dolls, 7 cars and 8 teddy bears.

- **a** Find the probability Laura takes at least one of each toy.
- **b** Find the probability Laura only takes teddy bears.
- **c** Given that she took exactly 2 dolls, find the probability she took exactly one car.
- **d** If Laura will only take teddy bears and dolls, find the probability she took exactly 3 teddy bears.

30 Roald is taking 4 books to read on his holiday. He has 3 biographies, 6 novels and 5 non-fiction books to choose from.

- **a** Find the probability Roald will take at least one biography.
- **b** Given that Roald takes at most one novel, find the probability that he takes exactly one non-fiction book.

31 Christa is painting 3 bedrooms in her house. Christa has 6 colours to choose from for the 3 bedrooms.

- **a** Find the probability that all the bedrooms are different colours.
- **b** Find the probability that at least 2 bedrooms are the same colour.
- **c** Given that at least two of the bedrooms are the same colour, find the probability that the third bedroom is a different colour.

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32 At a dessert buffet, a family chooses 5 different dessert plates. There are 6 bowls of ice cream, 3 bowls of pudding and 5 slices of cakes to choose from.

a Given that the family chooses exactly two bowls of ice cream, find the probability that only one bowl of pudding was chosen.

b Given that the family will choose more ice cream than cake and no pudding, find the probability that the family chose 4 bowls of ice cream.

33 A tour guide is taking groups on holiday and wants to mix up the nationalities of the groups, but she does this at random. Each group has 6 people. On this tour, there are 10 Australians, 7 Americans and 12 Chinese people.

a In one group, find the probability that there are 2 people of each nationality.

b Given that there are exactly 2 Chinese people in the group, find the probability that there are at least 3 Americans.

c Find the probability that there are no Australians in the group given that there are exactly 2 Americans.

34 The local school board needs 5 parents, 2 teachers and 3 local business owners to form a committee. Applications come in from 10 parents including Mr and Mrs Jones, 4 teachers and 6 business owners, including the local baker.

a Find the probability that Mrs Jones is on the committee if Mr Jones is on the committee.

b Given that the local baker is on the committee, find the probability that Mr or Mrs Jones (but not both) are also on the committee.

c Given that Mrs Jones won't be on the committee without Mr Jones, find the probability that they're both on the committee.

35 A teacher is creating a maths test with 10 questions. She has 12 calculus questions, 11 algebra questions and 7 trigonometry questions to choose from.

a Given that exactly 6 of her questions are algebra questions, find the probability that there are more calculus questions than trigonometry questions.

b Given that there are no trigonometry questions, find the probability that there are at most two calculus questions chosen.

36 A cricket team consists of 11 players, including a captain and a vice-captain. The coach decides on the basis of fairness to select the captain and vice-captain at random.

a How many different selections of captain and vice-captain are possible?

b Find the probability of a particular team member becoming captain or vice-captain.

c If the coach adopts a policy whereby former captains and vice-captains are not eligible to be reinstated in leadership positions, and if there are 4 members who have served as captains or vice-captains, find the probability of a particular member from the rest of the team becoming captain or vice-captain.

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37 10 Members of Parliament, including the Prime Minister and the Treasurer, have a row of 10 seats randomly designated for them at the party campaign launch.

a Find the chance that the Prime Minister is seated at either end of the row and the Treasurer is seated next to the Prime Minister.

b Find the chance that the Prime Minister is seated at least one seat away from either end of the row and the Treasurer is seated next to the Prime Minister.

38 A board of 12 company directors are having a board meeting at a circular table for 12. Find the probability that two particular directors, Luke and Eileen, are:

a Sitting next to each other.

b Sitting at least one seat apart.

c Sitting at least two seats apart.

d Sitting at least three seats apart.

e Seated such that Luke is sitting immediately to the right of Eileen.

f Seated such that Luke is sitting immediately to the left of Eileen

39 Luke and Maria randomly select places to sit at a particular table. The table has one seat placed on each side. Find the probability that they sit next to each other if the table is:

a Triangular (3 sides).

b Square (4 sides).

c Pentagonal (5 sides).

d k-sided.

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